

NetraSetu — ML Benchmarks (National Round, final)

PS 26038 · Team Game Of Codes

Every number below is measured, and each one states its **population, sample size and interval**. The canonical source for every classifier, conformal, external-validation, lesion and rule-engine figure is the ML lead's *ML Layer Final Report & Claims Ledger* (2026-09-21). Pipeline-level figures (smoke test, parity, soak test) come from the integration pass that followed it.

How to read this document. Section 1 holds the numbers we stand behind. Every section after it adds context: how the numbers degrade on a camera the model has never seen, what the deployed code path does on IDRiD's official test split, and what is still unproven. Where a number flatters us for a known reason, the reason sits next to the number.

Served model versions:

- classifier: `branchA_v2c` , conformal policy `ordinal_mode_interval_stratified_v3` ;
- vessel segmentation: `vessel_unet_v1` ;
- optic disc/fovea localization: `localization_v1` ;
- hard-exudate segmentation: `hard_exudate` (formerly `bright_lesion`);
- red-lesion segmentation: `red_lesion_unet_v2` .

Checksums are in `docs/RELEASE.md` .

1. Headline — DR severity classifier (Branch A, `branchA_v2c`), in-domain

The model: EfficientNet-B0 at 512×512 with an ordinal-aware loss. It was trained on APTOS 2019 + IDRiD, with an EyePACS subset added to the 5-class loss (weight 0.5) and photometric domain augmentation.

1a. Held-out test set: 628 images (550 APTOS + 78 IDRiD)

This set was held out from every training and calibration step for all four model versions.

Metric	Result	SIH target
Quadratic-weighted kappa (QWK)	0.884	—
Referable-DR sensitivity (argmax grade ≥ 2)	92.7%	> 90% 
Referable-DR specificity (argmax grade ≥ 2)	92.4%	> 85% 
Grade-4 (proliferative DR) recall	57.4% (31/54)	—
Grade-1 (mild) recall	55.0% (n = 60)	—

1b. Live operating point: the calibrated referable threshold

How the live referral works. The referral decision is not the argmax grade. It is a calibrated threshold on P(grade ≥ 2), set at **0.3873** for v2c.

How it was validated. It was checked by 50-fold cross-fitting on the pooled validation + test population (**n = 1,161**). In cross-fitting, the calibration is always fit on other folds than the one being scored.

Metric (cross-fit, n = 1,161)	Result [95% CI]	SIH target
Referable sensitivity at the live threshold	95.0% [92.8, 96.6] (478/503)	> 90%  , lower bound clears
Referable specificity at the live threshold	91.0% [88.6, 93.0] (599/658)	> 85%  , lower bound clears

These are in-domain numbers. They hold on the camera families the model was trained and calibrated on. Section 3 shows what happens on a camera it has never seen, and that section is part of the result, not a footnote.

1c. Why grade-4 recall is 57% and that is still safe

A proliferative case graded 3 instead of 4 is still referred: referral fires on the calibrated $P(\text{grade} \geq 2)$, not on the exact grade. In the cross-fit safety study (Section 2), **zero grade-4 cases were ever auto-cleared** across 1,000 fold assignments.

The low recall is about exact-grade resolution at the top of the scale, not missed referrals. Any grade-4 prediction by the CNN also forces full manual review (Tier C).

Thin-evidence note on the IDRiD-only slice. On the 78 IDRiD images within the held-out set, v2c's grade-4 recall is 0.600 (6/10) versus v1's 0.300 (3/10). With $n = 10$ grade-4 cases, this is too few for a strong claim either way; always quote it separately, never folded into the pooled number.

1d. Model lineage (same 628-image held-out set)

	v1 (384px)	v2a (512px)	v2b (+EyePACS, binary head)	v2c (deployed)
QWK	0.869	0.873	0.885	0.884
Referable sens (argmax)	86.0%	90.1%	84.9%	92.7%
Referable spec (argmax)	93.8%	93.8%	94.7%	92.4%
Grade-4 recall (n = 54)	0.444	0.537	0.463	0.574

Disclosed deviation from our own promotion rule. The pre-declared rule (paired AUC gain on an untouched external half) selects v2b: v2c's AUC gain over v2b was $-0.009 [-0.029, +0.012]$, crossing zero. We deployed v2c anyway, on a more decision-relevant metric: how often the auto-clear tier is wrong about a referable case on an unseen camera. That rate was 2.3% for v2c versus 12.8% for v2b (Section 3). We made this call with both models' full external numbers in hand, and it is stated here rather than hidden.

2. Safety — conformal confidence tiers (in-domain, cross-fit)

Method. Class-conditional (referable-stratified) conformal prediction decides which cases may skip human review (Tier A). It uses two strata: grades 0–1 and grades 2–4, with $\alpha = [0.30, 0.05]$. The guarantee is therefore strongest on referable cases.

Setup. 50-fold cross-fit on the pooled population, $n = 1,161$, deployed model v2c.

Guard	Threshold	Result
Referable-stratum coverage, lower confidence bound	$\geq 93\%$	94.2% 
False auto-clear rate, true referable cases (upper bound)	$\leq 5\%$	0.0% 
False auto-clear rate, true grade ≥ 3 cases (upper bound)	$\leq 2\%$	0.0% 
Mean prediction-set size	≤ 2.5	2.01 
Grade-4 cases auto-cleared, across 1,000 fold assignments	—	0
Tier distribution (pooled)	—	A 38.4% · B 43.8% · C 17.8%

How the method evolved. Don't describe the current method as the original one.

- 1. **Legacy marginal LAC.** A single global threshold that could produce non-contiguous or empty prediction sets: 3.8% non-contiguous on the 628-image test set.
- 2. **Mondrian v2.** Per-grade calibration. An internal sanity gate caught a scoring defect before shipping: confident, correct predictions received the highest nonconformity score, which saturated the quantile at about 0.996.
- 3. **v3 (shipped).** Corrected scoring, with zero nonconformity at the predicted grade that grows monotonically away from it, plus referable stratification. It is validated by 90/90 automated tests: golden vectors plus 10,000-trial property tests, with exact MATLAB/Python agreement.

Structural note. Two safety clauses exist in code but never change an outcome today. Both are subsumed by the calibrated threshold, so they are not described as active safety layers.

- **Referable threshold demotes Tier A → Tier B:** Tier-A eligibility already implies $P(\text{grade} \geq 2)$ well below 0.3873.
- **$P(\text{grade } 3) + P(\text{grade } 4) > 0.5 \Rightarrow \text{referable}$:** any such case already exceeds the 0.3873 threshold on $P(\text{grade} \geq 2)$.

What "guaranteed" does and doesn't mean. Conformal coverage assumes the test population resembles the calibration population. On an unseen camera it undershoots (Section 3). The accurate phrase is: *class-conditional calibrated confidence, validated in-domain, with a measured and disclosed degradation under camera shift.*

3. External validation — a camera the model has never seen (Messidor-2)

The dataset. Messidor-2: 1,744 gradable images from 874 patients, a Topcon camera, and independently adjudicated grades. It was never used for training, calibration or threshold selection.

The split. Patients were split once by patient-ID parity, before any numbers existed. The even half chose between model candidates. The **odd half, n = 872, was never touched until one audited final run**, and every figure below comes from that report half.

- Intervals are 95% patient-level bootstrap.
- False-auto-clear rates use exact Clopper–Pearson bounds.

Metric	In-domain	Unseen camera (Messidor-2 report half, n = 872)
AUC of $P(\text{grade} \geq 2)$	~0.98	0.924 [0.900, 0.945]
Referable sensitivity at the shipped threshold	95.0%	75.2% [68.6, 81.4]
Referable specificity at the shipped threshold	91.0%	93.9% [91.8, 95.8]
QWK	0.884	0.717
Grade-4 recall	57.4%	46.2% (n = 13)
False auto-clear, true referable (final Tier A)	0.0%	2.3% (5/218), CP95 [0.8%, 5.3%]
False auto-clear, true grade ≥ 3	0.0%	0.0% (0/45), CP95 [0%, 7.9%]
Tier A / B / C share	38.4 / 43.8 / 17.8%	36.5 / 22.0 / 41.5%

What this supports, stated plainly:

- **The >90% sensitivity target does not hold on a camera the system hasn't been validated on.** At the shipped threshold, sensitivity is 75.2%. Specificity holds up fine.
- **Ranking quality is the limit, not calibration.** Refitting the calibration on up to 437 local patients never closed v2c's residual coverage gap (it peaks around 87.6% against a 92.4% target).
- **The auto-clear tier stays close to its safety guard under shift.** The rate is 2.3%, with an upper bound of 5.3%, so read it as *close to the 5% guard*, not a clean pass. No severe case (grade ≥ 3) was auto-cleared.

- **The policy consequence.** No case from a camera or site that hasn't been locally validated may auto-clear; it is capped at Tier B (AI-assisted human review) with reason `unvalidated_camera` . This is the project's answer to domain shift. It is not a claim that the model generalizes; it is a control that stops the model auto-clearing when it might not.
- **Framing for any audience:** "AI triage with mandatory human review on cameras that have not been locally validated", not "90%+ sensitivity anywhere."

4. Deployed code path on IDRiD's official test split (smoke test, not a validation)

What it is. An independent run of the full production grading path on the 103 official IDRiD test images (`scripts/audit/eval_idrid_test.py`).

Metric (n = 103)	Result
QWK	0.841
Referable sensitivity, live referral flag	100% (64/64)
Referable specificity, live referral flag	82.1%
Grade-4 recall	13/13
Tier A / B / C	4 / 70 / 29 — no Tier-A case was referable

Do not quote this table as accuracy. The temperature scaling and referral threshold were fit on a pool that includes these 103 images, and the set is 62% referable, which moves both sensitivity and specificity. What it establishes is narrower but real: the integrated pipeline, run end to end, grades consistently with the model's validated behavior and silently under-grades nothing on this set. For accuracy, use Sections 1–3.

5. Segmentation, localization and lesion models

Model	Test data	Metric	Notes
Vessel U-Net (<code>vessel_unet_v1</code>)	CHASE_DB1, held out	Dice 0.777	Not retrained this round
Same model, no retraining	DRIVE (unseen dataset, n = 20)	Dice 0.619	Real cross-dataset gap. The live threshold is a fixed 0.5; the designed domain-adaptive threshold was never implemented
Optic disc / fovea localization (<code>localization_v1</code>)	IDRiD	Mean error 16 px (disc) / 32 px (fovea) at native resolution	
Hard-exudate U-Net	IDRiD	Dice 0.583 per image / 0.733 global	
Red-lesion U-Net v2, deployed (3-class: background / MA / haemorrhage)	IDRiD val, n = 16	Dice 0.599 merged · MA 0.442 · haemorrhage 0.571	Tuned and evaluated on the same 16 images, so this is thin evidence
Red-lesion U-Net v1 (single merged class)	same 16 images	Dice 0.535	Predecessor

Red-lesion v2. Its merged Dice overlaps v1's within statistical intervals, so v2 is **not a proven accuracy gain**. What it adds for certain is real, separate microaneurysm and haemorrhage counts per quadrant. It also brings class-specific minimum-area filters: MA 5 px (component F1 0.648) and haemorrhage 10 px (F1 0.595). A single 10 px floor would have discarded most true microaneurysms, whose median ground-truth size is 6 px across 3,452 components.

Fovea reliability gate. When the fovea heatmap has no real peak, the case is flagged `foveaUnreliable` (threshold 0.37 in raw heatmap units, not a probability).

- **Catch rate:** it catches both known gross localization failures in the held-out split. With only 2 positives, the interval is wide: 15.8–100%.
- **False alarms:** 5.3% [1.5, 12.9] on otherwise-good localizations.
- **Flag rates:** 2.7% of IDRiD images and 14.6% of Messidor-2 images.

Not detected, by design. Cotton-wool spots (soft exudates) have no detector, because the pixel-level training data is too sparse. The field is always reported as unmeasured, never as a false zero. Venous beading and IRMA detectors were not built for the same reason.

Neovascularization (NV) suspicion score: built, tested, failed. This was a vessel density, tortuosity, fractal-dimension and branching proxy near the optic disc.

	IDRiD test (n = 103, 13 PDR)	Messidor-2 (n = 1,744, 35 PDR)
AUC, PDR vs. all other grades	0.286	0.379

Both results are below chance. The cause was investigated: the vessel model, trained on healthy eyes, segments *less* vessel signal on diseased retinas, which swamps the NV signal. The score is **not wired into any decision**. Grade-4 cases are caught by the classifier, by the CNN-grade-4 → Tier C rule, and by mandatory review on branch disagreement.

6. Rule engine (Branch B — explicit ICDR "4-2-1" criteria)

What it is. Plain, auditable code on quadrant-mapped lesion counts, with no learned weights. Its frozen thresholds are `RED_FLOOR=3` , `GRADE3_QUAD_MIN=3` and `RULE_MAX_GRADE=3` . It is deliberately capped at grade 3, since there is no validated NV signal (Section 5).

Exact agreement with ground truth, IDRiD official test (n = 103)	Result
Original thresholds, v1 lesion counts	58.3%
Recalibrated thresholds, v1 counts	57.3%
Recalibrated thresholds, v2 counts (deployed)	60.2%

These three overlap within their intervals. An earlier 71.4% figure was measured on 14 hand-picked images, is **superseded**, and must not be quoted.

Why a 60% rule engine is still useful. Branch B's job is not to beat the CNN. It gives an independent, human-auditable second opinion: agreement strengthens a case's tier, and **disagreement unconditionally forces mandatory review** with an explicit grade chosen by the ophthalmologist. We report no system-wide Branch A/B agreement rate, because the only candidate file is a curated set of debugging examples, not a representative sample.

7. Engineering integrity

Check	Result
PyTorch ↔ ONNX ↔ MATLAB tensor parity, all 9 model artifacts	Max difference 1×10^{-6} to 4×10^{-5} (threshold 0.01)
Calibrated classifier, Python vs. MATLAB inference, 10 real IDRiD images	Grade and tier agree 10/10; max probability difference 5.1×10^{-7}
Red-lesion v2, ONNX vs. MATLAB	Max tensor difference 1.4×10^{-5} ; MA/haemorrhage counts identical on all 10 parity images
Full-pipeline soak test: every IDRiD image on disk through capture → quality gate → sync → grading	447/447 graded, 0 failed, 0 timed out (throughput and reliability, not accuracy)
End-to-end central grading latency per case (warm MATLAB session, dev laptop: Ryzen 7, RTX 4050, 24 GB)	p50 10.9 s / p95 22.2 s (n=10) , measured via <code>POST /api/v1/cases</code> → <code>poll /status</code> until <code>graded</code> , 10 real submissions through the live running stack. With n=10, "p95" is effectively the max of 10 samples, not an interpolated percentile — treat it as an upper bound seen, not a statistically tight bound. Range was 10.9s–43.1s; the two outliers (23–43s) look like queue/scheduling variance under a cold-ish run, not a different code path.
Local quality-gate latency (desktop, MATLAB)	p50 0.49 s / p95 3.2 s (n=10) , measured via 10 direct <code>qualityGateMain()</code> calls in a live MATLAB session (warm toolboxes, not counting MATLAB's own interpreter startup). Range 0.17s–3.2s.
Quality gate, JS implementation vs. MATLAB reference	Agreement within ~0.005 on reference values

Automated tests:

Suite	Result
Conformal (MATLAB + Python)	90/90
Fovea gate	12/12
PHC backend	41/41
Mobile	28
Central backend	No automated suite; targeted verification scripts only

8. Limitations, stated on purpose

- In-domain vs. unseen camera.** Sensitivity is 95.0% in-domain but 75.2% on an unseen camera at the shipped threshold, and local recalibration does not close v2c's gap. That is why unvalidated cameras cannot auto-clear.
- External auto-clear margin.** The false-auto-clear rate on the unseen camera is 2.3%, with an upper bound of 5.3%: close to the 5% guard, not a clean pass.
- Top-grade resolution.** Grade-4 exact recall is 57.4% in-domain. It is safe because of referral thresholding and Tier C routing, but the top-grade resolution is a real limitation.
- Promotion-rule deviation.** v2c was deployed against our own promotion rule's pick (v2b), for a disclosed safety reason.
- Red-lesion evidence is thin.** v2 was evaluated on 16 images, the same ones it was tuned on.
- Neovascularization failed validation** and is not used. Cotton-wool spots, venous beading and IRMA are out of scope.
- Vessel cross-dataset gap.** Dice is 0.777 in-domain vs. 0.619 on DRIVE, with a fixed 0.5 threshold.
- Rule engine.** Exact agreement is 60.2% on n = 103; its value is the auditable second opinion and the disagreement trigger, not standalone accuracy.
- No clinical data.** No real-patient or prospective data; all results are on public datasets.
 - Two training datasets are Indian populations: APTOS (Aravind Eye Hospital) and IDRiD (Nanded, Maharashtra).

- No subgroup analysis (age, sex, camera model within a dataset) has been done: the public datasets don't carry the demographic fields needed.

10. **DME.** Diabetic macular edema cannot be reliably ruled out from colour fundus photographs alone.
11. **Quality gate not validated against labels.** Its thresholds are engineered heuristics, checked for MATLAB/JS agreement but not validated against a gradability-labelled dataset.

Reference point, not a comparison. IDx-DR's FDA pivotal trial reported 87.2% sensitivity and 90.7% specificity for referable DR (900 patients, 10 sites). IDx-DR is autonomous; NetraSetu is decision support with an ophthalmologist confirming every positive.

9. Superseded figures — do not quote

These appeared in earlier drafts, the internal-round deck, or working notes. They are replaced by the figures above; anyone presenting NetraSetu should use the right-hand column.

Old claim	Why it's retired	Use instead
"100% referable sensitivity, 13/13 grade-4" (IDRiD official test, n = 103)	Calibration pool includes these images	95.0% / 91.0% (cross-fit, n = 1,161); grade-4 57.4% (n = 54)
QWK 0.883 / 0.869, sensitivity 86%, specificity 94–96%	Internal-round model v1 (384px)	v2c: QWK 0.884; 95.0% / 91.0% at the live threshold
"Grade-4 recall 0.444"	v1, pooled; always needs population and n	v2c 57.4% (31/54) pooled; IDRiD-only 6/10
"71.4% rule-engine agreement"	n = 14, hand-picked	60.2% (IDRiD official test, n = 103)
"14.6% of prediction sets non-contiguous"	Source never confirmed	3.8% (legacy method, n = 628)
Vessel Dice 0.80 (CHASE_DB1, n = 28)	n = 28 is the whole dataset, training images included	0.777 held out
"Guaranteed confidence"	Coverage only holds in-domain	"Calibrated confidence, validated in-domain, with disclosed degradation under camera shift"
">90% sensitivity" with no population	False on an unseen camera	95.0% in-domain; 75.2% on an unseen camera
"50–100 local images recover the sensitivity gap"	True for v2b, not for the deployed v2c	Unvalidated cameras are capped at human review
Lesion detection "evaluated with FROC"	No FROC analysis was run	Dice and component-level F1 (§5)
Domain-adaptive vessel threshold (0.5 / 0.10)	Designed, never implemented	Fixed 0.5 threshold

10. Sources and reproduction

Every figure traces to one of these. Model binaries are distributed separately and verified by checksum (`npm run models:verify`).

Figures	Source
\$1–\$3, \$5 (lesion, fovea, NV), \$6, \$9	ML Layer Final Report & Claims Ledger, 2026-09-21 — not in repo . No file matching this name/date exists anywhere in this checkout (searched by name and by date); it exists only outside this repository (the ML lead's own copy). Cross-checked instead against <code>diagnostics/MODEL_INTERFACE_REFERENCE.md</code> and <code>models/red_lesion_v2_metrics.json</code> , which reproduce the same figures — see the \$5 row below.
\$2 cross-fit guards, tiers, grade-4 auto-clears, Python/MATLAB parity	<code>central-system/backend/ml-pipeline/diagnostics/out/conformal_v3_crossfit_report_branchA_v2c.json</code>
\$3 external validation	<code>central-system/backend/ml-pipeline/docs/messidor2_v2c_final_external_report.md</code> , <code>central-system/backend/ml-pipeline/docs/messidor2_v2b_final_external_report.md</code> (path corrected — these are nested under <code>ml-pipeline/docs/</code> , not the top-level <code>docs/</code>)
\$4 smoke test	<code>scripts/audit/eval_idrid_test.py</code>
\$5 fovea gate	<code>central-system/backend/ml-pipeline/diagnostics/out/fovea_gate_report.json</code>
\$5 segmentation	Vessel 0.777: <code>diagnostics/MODEL_INTERFACE_REFERENCE.md:257</code> (CHASE_DB1, subject-level 22-train/6-val split, seed 42 — the genuine held-out score; <code>diagnostics/out/m2_metrics.json</code> 's 0.8024 scores against all 28 images including training subjects and is not the same measurement). Hard-exudate 0.583/0.733: <code>diagnostics/MODEL_INTERFACE_REFERENCE.md:340,343</code> (original 43-train/11-val IDRiD split; per-image mean / pixel-pooled). Red-lesion v2 0.599/0.442/0.571: <code>models/red_lesion_v2_metrics.json</code> (<code>best_val_dice_merged</code> , <code>val_dice_ma</code> , <code>val_dice_he</code>). Red-lesion v1 0.535: <code>models/red_lesion_predictions(model5)/red_lesion_metrics.json</code> (<code>best_val_dice</code> , n=16) — also independently reproduced in <code>diagnostics/out/m45_metrics.json</code> 's <code>m5_heldout.per_image_dice</code> (0.53508) on the same 16 images. Localization 16 px / 32 px: source not located. The closest file, <code>diagnostics/out/m3_metrics.json</code> (heldout, n=77), reports different numbers (mean 26.9 px / 85.6 px native, median 21.2 px / 31.2 px native) — neither matches 16/32 exactly. Flag this specific figure to the ML lead for its source before publishing it; do not treat it as reconciled.
\$7 artifact parity	<code>central-system/backend/ml-pipeline/diagnostics/out/artifact_manifest.json</code>
\$7 soak test	<code>scripts/test-full-dataset.js</code>